Jihun Rho

GPU Performance Engineer

Computer Vision Engineer

B.S. Electronic Engineering, Sejong University

CUDA · Computer Vision Real-Time Systems

CONTACT

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GitHub profile

Portfolio website

CORE SKILLS

CUDA / GPU

C++, CUDA kernels, memory tuning,

stream overlap, occupancy

Profiling

Nsight Systems, Nsight Compute, performance analysis

Computer Vision

OpenCV, image processing, detection, tracking

Inference

TensorRT, YOLO, camera-radar fusion, real-time evaluation

Systems

C/C++, Multithreading, Linux, Windows

Robotics

Robot control, REST, TCP/IP, Modbus, Digital I/O

EDUCATION

B.S. in Electronic Engineering — Sejong University | 2018

ABEEK-accredited engineering program

PROFILE

GPU Performance Engineer | Computer Vision Engineer

GPU Performance Engineer with **4 years and 8 months of hands-on CUDA/C++ experience**, specializing in **GPU kernel optimization**, **high-performance image processing**, and **industrial machine vision**. Bringing **7 years and 6 months** of combined industry and research experience spanning computer vision, deep learning, and robotics.

Portfolio: **931njhthe-star.github.io**

EXPERIENCE

Research Engineer — High-Throughput Image Processing Pipeline Development & Optimization

Enclony — Advanced Technology Research Institute | Mar 2021 – Oct 2025

- Developed and optimized **high-performance CUDA/C++ machine-vision software** for industrial pharmaceutical inspection systems.
- **Redesigned image-processing pipelines** with selective CUDA acceleration for compute-intensive workloads, retaining efficient CPU processing for lighter tasks and optimizing CPU–GPU workload distribution.
- Achieved **11.2×** higher image-processing throughput than a single-core CPU baseline and **approximately 40%** higher throughput than a multi-core CPU-only implementation, reaching **2.52 million images per hour**.
- Processed **10 RGB images (640×480) and 2 3D camera-derived height maps** per object for multi-view surface inspection.

Researcher — Deep Learning & Computer Vision

Sejong University — Intelligent Vehicle Recognition Systems Laboratory | Nov 2019 – Oct 2020

- **Vehicle Detection:** Evaluated **YOLOv3/v4 and MobileNet-SSD** on rooftop-captured video footage. Curated tens to hundreds of thousands of extracted frames, managed JSON/text annotations, and performed detection evaluation and error analysis.
- **Traffic Sign Recognition:** Developed a **3D-to-2D projection simulation** using camera geometry and coordinate transformations. Evaluated LeNet and SVM recognition performance for traffic signs rotated around different 3D axes, analyzing the effect of rotation angles on accuracy.

Software Engineer — Robotics & Vision Systems

INA Oriental Motor — Technology Research Center | Jan 2018 – Oct 2019

- **Planned and integrated an industrial robotics demonstration system** combining autonomous mobile robots, six-axis robots, 3D vision systems, and pneumatic grippers.
- **Developed integration software** using Python, REST APIs, TCP/IP, Modbus, and digital I/O, implementing monitoring, motion coordination, and gripper-control workflows across devices.
- Integrated **deep-learning-based 3D object recognition and random bin-picking** into robotics demonstrations.

PROFESSIONAL DEVELOPMENT

Multi-Agent AI Orchestration Program
SK Networks Family × Encore AI Campus | Aug 2026 – Jan 2027 (Expected)

Earlier Training (2016–2019)
Practical Machine Learning · Python Web Programming · Arduino Electronics · Security & Network Administration

SELECTED PROJECTS

Applied work across perception, robotics integration, and AI systems.

DriveFusion — Real-Time Camera–Radar Perception & Evaluation
View project page

Independently built a CARLA-based system combining **YOLO11n inference** through a C++/CUDA TensorRT module, **vehicle tracking**, **radar-assisted distance estimation**, and a Qt monitoring dashboard. Across nine weather and time-of-day test conditions (16,491 common samples), radar-assisted tracking reduced overall distance MAE by **5.41%**; the reduction reached **39.42%** on 437 radar-corrected samples.

Technologies: C++, CUDA, TensorRT, YOLO11n, Qt, CARLA, radar fusion

Robotics & 3D Bin-Picking System Integration
View project page

A **professional 3D bin-picking integration project** connecting a MiR AMR, Doosan six-axis robot, and Solomon 3D vision, presented through publicly available demonstration materials.

Technologies: Robot control, 3D vision, REST API, TCP/IP, Modbus, digital I/O

ADDITIONAL PROJECTS

SubSync — YouTube English-Learning Extension View GitHub repository

Built an AI service with FastAPI, Supabase, Redis, and monitoring.

CareerLens — AI Resume & Job Matching View GitHub repository

Designed a parallel LangGraph pipeline and integrated the service backend.

Speak Up — English Academy Website Visit live website

Built and deployed an academy website independently; now in active use.

Technologies: Python, FastAPI, LangGraph, PostgreSQL, Redis, React, TypeScript

SELECTED AWARDS

Excellence Award — 2nd ICT CoC Hackathon (2020) | NIPA & Seoul Metropolitan Government

Awarded for developing a YOLOv3 aerial detection system for unauthorized buildings and rooftop gardens.

Technologies: C++, YOLOv3, deep learning, computer vision, object detection

Additional Awards (2016–2017)

Bronze Award, Sejong University Creative Design Competition (2017) · 4th Place, Sejong University Software Competition (2016) · Outstanding Talent Award (2016)